Code No: 138AC **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, June - 2024
ADVANCED ALGORITHMS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Define dynamic programming.	[2]
b)	Discuss the role of algorithms in computing.	[3]
c)	Define greedy method.	[2]
d)	Discuss about maximum flow algorithm.	[3]
e)	What is linear equation?	[2]
f)	Give a brief summary on merging network.	[3]
g)	What is the best case of Naive algorithm for pattern searching?	[2]
h)	What is the average and best-case running time of the Rabin-Karp algorithm?	[3]
i)	What is non-deterministic algorithm?	[2]
j)	Discuss about vertex cover problem.	[3]

PART – B

(50 Marks)

2. Compute the following sequences of keys in ascending order with quick sort method 21,54,34,10,91,70,50,65,53,23,33. [10]

OR

3. Give an example to explain in detail matrix chain multiplication using dynamic programming. [10]

4. What is Topological sorting? Illustrate topological sorting with algorithm. [10]

OR

5. What is minimum cost spanning tree? Write algorithm for Kruskal's technique with Illustration. [10]

6. Discuss in detail bitonic Sorting Networks with an example. [10]

OR

7. Explain in detail about the representation of inverting matrices. [10]

8. Describe briefly about Knuth-Morris-Pratt algorithm with a suitable example. [10]

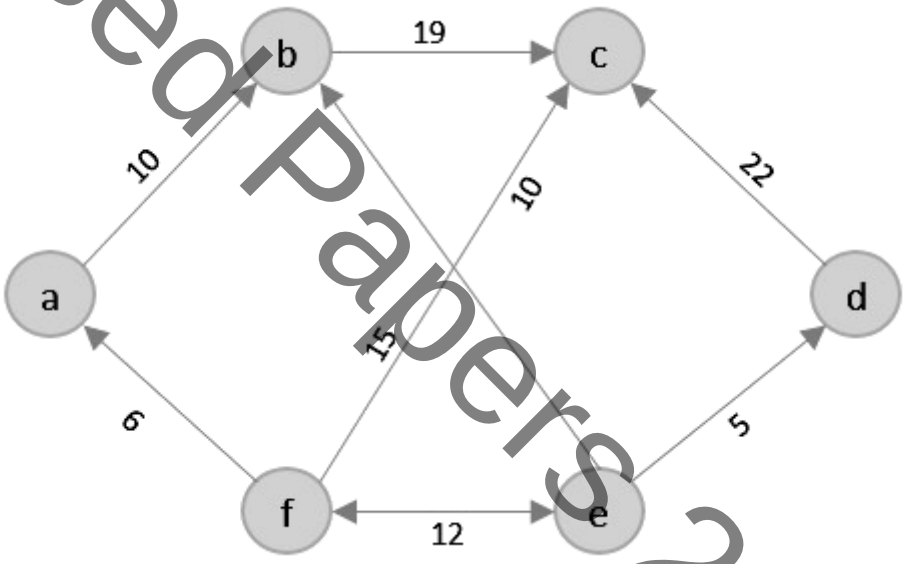
OR

9. Discuss in detail about matching of a string using finite automata. Give example. [10]

10.a) Explain about Polynomial time and polynomial time verification.
b) Distinguish between NP-hard and NP-complete. [5+5]

OR

11. Write an approximation algorithm for Travelling Sales person problem and implement for the following graph. [10]



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